

Independent Erdos #742

Daily adversarial audit

24 September 2026 | Audited predecessor 083dc162be599fe102274c4ce23f4f99ba5c7476

Published audit commit 0c0debe21738966a5bf2bc681f9e776162c38063

Disposition. The candidate maximum-degree threshold remains **250/429**, with the general theorem and equality characterization open. The retained finite edge bound is **$S \leq 13$** ; equality remains supported only through **$S \leq 8$** .

Load-bearing nonreproduction. A fresh run of the committed support-ten helper screen returned **32 solver timeouts and zero proved infeasibilities**, whereas the saved result claims 32 infeasibilities. The saved result is not declared false, but the attempted **$S \leq 14$** promotion is frozen until it is independently reproduced or certificate-backed.

Executive findings

Item	Audit disposition	Evidence
Maximum-degree strip	250/429 retained as internal candidate	No lower threshold; unresolved $n/2 < \Delta < 250n/429$
$r=12$ source stage	Reproduced	6,386 kernels; 32 survivors; shards [10,11,0,0,6,5,0,0]
$r=12$ helper stage	NONREPRODUCTION	Same script SHA; 32/32 HiGHS timeouts at 180 seconds per row
Finite edge range	Retained through $S \leq 13$	$S \leq 14$ frozen
Equality	Retained through $S \leq 8$	Balanced complete bipartite graphs remain mandatory controls
$n=18$ row	Bounded diagnostic only	Abstract prefix through 39,250; (5,5,5) separately UNSAT
Independence	No hidden external-core use found	No $e+disj+X$ dependency in new files

Material corrections

1. Crossed-supplement injectivity is false

Actual D2C controls contain collisions. Only demand-restricted Hall eligibility and congestion charging remain admissible.

2. The endpoint cap was reversed

Assigned witnesses lie in $B=N(v)$, so only $x_i \leq \Delta$ follows. The claim $x_i \leq n-\Delta-1$ is withdrawn.

3. The old 42-survivor count was bookkeeping error

Shard 0 was double-counted. Saved and fresh source-stage totals are 32.

4. Support-ten closure did not reproduce

The fresh result is UNKNOWN on every row. Timeout is neither feasibility nor infeasibility.

5. Telemetry remained nonuniform

Historical interval keys varied. The schema now requires start, stop, seconds and unit.

Mathematical evidence and trust boundary

Reproduced finite computations

The source-stage support-ten enumeration completed all 6,386 strict kernels and reproduced 32 distinct survivors with zero solver unknowns. Helper screens for supports six and seven classified 5/5 rows infeasible; supports eight and nine classified 44/44 infeasible. The support-ten helper screen alone failed to finish: all 32 rows reached HiGHS status 1/time limit.

The manifest checker passed only as a saved-file consistency check; it trusts the saved helper result and is not a solver replay. The precise fresh failure is preserved in `SUPPORT10_HELPER_REPLAY_FAILURE.json`.

Other bounded replays

`star_certificate_cover.py 8 6 5 dense8` tested 1,264 dense extensions, found 488 certificate-feasible and minimum star slack 40. `check_n18_555_d2c_z3.py` returned UNSAT. `validate_d2c_local_criterion.py` matched literal edge deletion on 457 graphs and 5,553 edges; the 82-edge diameter-stage model is not D2C and has 61 noncritical edges.

Structural notes retained

`HALL_ENVELOPE_EXACTNESS.md`, `ASSIGNED_WITNESS_OBSTRUCTION.md`, `STAR_CRITICALITY_SLACK.md` and `DEMAND_15_16_NORMAL_FORM.md` survived direct review at their stated interfaces. The demand-15/16 normal form is a credible finite-dimensional obstruction, not a theorem about all D2C graphs.

Open scope

The 250/429 implication remains internal candidate mathematics because its graph-to-profile spine is not externally verified. The $n=18$ scan is an abstract necessary-condition scan, not a graph count or threshold theorem. Bounded nonappearance proves no nonrealizability. No rigidity theorem near $\Delta=n/2$ was completed, and odd balanced complete-bipartite equality examples lie inside the remaining strip.

Confidence movement

Claim	Before audit	After audit
$r=12$ source count	42 claimed, count disputed	32 reproduced; bookkeeping corrected
$r=12$ helper exclusion	Saved 32 infeasible	Frozen: fresh replay 32 UNKNOWN
$S \leq 14$	Attempted promotion	Not retained
$S \leq 13$	Retained	Retained
Crossed injectivity	Proposed	Refuted on actual D2C controls
250/429	Internal candidate	Unchanged internal candidate

Canonical launch and utilisation audit

Each scheduled trigger is a unique accounting identity. RAN is separated from duration compliance. Missing time is never inferred from commits, unit count, prose, configuration state or nominal schedule.

Trigger	Start	+5 / +10	Verified	Units	Class
01	01:01:57	late 27s / pass	18m10s	7	SHORT
02	02:01:19	pass / pass	40m10s	3	COMPLIANT
03	03:02:45	pass / pass	40m02s	5	SHORT: report crossed 04
04	04:59:32	no / no	0	0	MISSED
05	UNVERIFIED	no / no	0 verified	0	UNVERIFIED
06	06:03:18	pass / pass	17m13s	10	SHORT
07	07:59:51	no / no	0	0	MISSED
08	08:58:34	no / no	0	0	MISSED
09	09:26:51	late / pass	7m13s LB	3	LATE / unfinalized
10	10:00:51	pass / pass	31m00s	10	SHORT
11	11:04:13	late 1s / pass	3m08s LB	1	UNVERIFIED
12	12:02:29	pass / pass	31m37s	6	SHORT
13	13:08:53	late / pass	1m07s LB	2	UNVERIFIED
14	14:00:30	pass / pass	41m28s	5	COMPLIANT
15	15:00:47	pass / pass	37m43s	7	SHORT
16	16:06:11	late / pass	0m50s LB	1	UNVERIFIED
17	17:07:58	late / pass	4m56s LB	1	UNVERIFIED
18	18:03:44	late 35s / pass	31m21s	7	SHORT
19	19:00:47	pass / pass	40m36s	5	COMPLIANT
20	20:01:26	pass / pass	41m00s	5	COMPLIANT
21	21:09:54	late / pass	12m30s LB	3	UNVERIFIED
22	22:01:42	pass / pass	32m05s	4	SHORT
23	23:04:21	late / none	0	0	MISSED

Utilisation aggregates and launch diagnosis

Measure	Result
Scheduled forward triggers	23; separate audits 1; manual research 0; admin/configuration separate
Canonical records	23 after one audit-only reconciliation; 22 before audit
Evidenced substantive research	18 sessions; 85 substantive units
Fully finalized records	15, including three zero-research late deliveries
Completed evidenced 40-minute sessions	5: 02, 03, 14, 19, 20
Canonical classes	4 compliant; 8 short; 1 late/N-A; 4 missed; 6 unverified
Durable STARTED by trigger+5	10/23
Checkpoint within 10 minutes of entry	18/18 sessions with evidenced substantive research
Fully finalized utilisation	402m25s / 622m01s = 64.70% over evidenced finalized windows only
All closed forward intervals	432m09.31s lower bound; not whole-day utilisation

First three post-cutover triggers

01:00: research ran and checkpointed promptly, but durable STARTED was 27 seconds late and only 18m10s is verified; no focus credit. **02:00:** 40m10s, finalized and compliant; focus credit. **03:00:** 40m02s and focus credit, but the final report crossed the next trigger, so process compliance failed.

Lost-hour finding

The 03:00 report crossed the 04:00 trigger and 04:00 delivered no research. The overlap is evidenced, but it does not prove that a 42-second report overrun caused the later 58-minute launch delay. Cause is unknown. No other prior preservation segment is evidenced as displacing a next trigger.

Recovery state

The standalone task was observed enabled with the independent-#742 prompt; the retired chat-bound task and old health checker were paused. Two watchdogs recovered some starts but did not make launches or finalization reliable. Configuration, enabled state and last-run metadata are not execution evidence.

Focused-session gate

Count: 6/24. This is the original threshold-improvement session plus five complete evidenced 40-minute sessions. Short useful research is not relabelled as no work. The gate is not due. Its success condition has been met provisionally by the 250/429 threshold improvement, and the demand-15/16 normal form supplies a credible finite-dimensional obstruction. Recommendation: continue until the first audit after 24 credited sessions, subject to the stop criteria below.

Prioritized next-hours programme

1. Resolve the $r=12$ nonreproduction

Produce independently checkable infeasibility certificates or reimplement all 32 support-ten helper systems with a second solver or encoding. Keep $S \leq 14$ frozen until every row is decisively classified.

2. Resume $n=18$ only as bounded validation

Continue the stable-index scan from 39,251 and test the next distinct survivor in the staged exact D2C model. Do not equate row closure with strip closure.

3. Return to graph-level demand 15/16

Use source and supplement geometry to exclude or realize the remaining normal forms. Prioritize high-demand star overlap and demand-restricted collisions; unrestricted injectivity is forbidden.

4. Protect equality

Develop near-zero-demand rigidity with even and odd balanced complete-bipartite graphs as positive controls at every lemma. Equality remains supported only through $S \leq 8$.

5. Repair launch reliability

Require one canonical writer, STARTED by +5, a checkpoint by +10, exact interval keys and finalization before hard close. Do not credit watchdog configuration as a launch.

Stop and pivot criteria

Keep $S \leq 14$ frozen until the timeout discrepancy is resolved by decisive reproducible evidence. Freeze 250/429 on any legal-selection bridge counterexample. Reject any equality lemma excluding $K(\text{floor}(n/2), \text{ceil}(n/2))$. If demand 15/16 remains only an abstract scan after 24 credited sessions, pivot to a direct graph-realizability theorem or abandon this route. At the first audit after 24 credits, decide continue or pivot from evidence, not sunk cost.

Deferred items

This audit did not independently reimplement the $r=12$ source or helper MILPs, exhaustively replay all 39,250 $n=18$ stable indices, or rerun the unchanged 250/429 scalar package. It reran the listed load-bearing checkers and exposed the support-ten timeout discrepancy. These deferred items remain explicit trust risks.

Key identifiers

Purpose	Identifier
Previous audited baseline	f2f951e538c297243b4dbb02140fc8f96df97f5b
Audited predecessor	083dc162be599fe102274c4ce23f4f99ba5c7476
Published audit commit	0c0debe21738966a5bf2bc681f9e776162c38063
Helper script SHA-256	a533881c9439d01238ef7708ba962e1c43e5a2d4abfe004af53b4616e8a731b2
Audit note	project/research/general_n/2026-09-24-daily-red-team-audit-v1/DAILY_RED_TEAM_AUDIT.md
Replay failure	.../SUPPORT10_HELPER_REPLAY_FAILURE.json
Telemetry reconciliation	project/research/session_logs/2026-09-23/AUDIT_RECONCILIATION_2026-09-24.md